

LECTURE 23: MATH1061/7861
TOPICS: COUNTABLE AND UNCOUNTABLE
SETS
DATE: THURSDAY APRIL 22, 2026

Learning objectives:

- (1) Understand the definition of a countable set and an uncountable set.
- (2) Learn how to prove statements about the countability of a set.
- (3) Note: Material of this lecture is not assessable.

↳ Won't be on Exercise sheet
or on the final exam.

Definition: A set is **countable** if there exists a bijection between the set and $\mathbb{N}$.

Consider a function $f : \mathbb{N} \rightarrow S$.

bijjective = injective + surjective

f injective means:

$$f(m) = f(n) \implies m = n.$$

f surjective means:

$$\forall s \in S, \exists n \in \mathbb{N} \\ \text{s.t. } f(n) = s.$$

$$\mathbb{N} = \{0, 1, 2, \dots\}$$

Example: $\mathbb{Z}$ is countable.

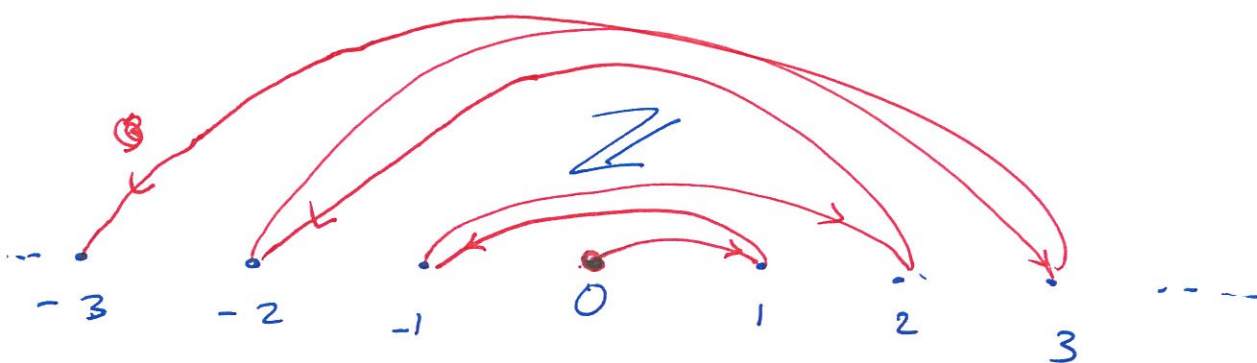
] Prove these.

② $\mathbb{N} \times \mathbb{N}$ is countable

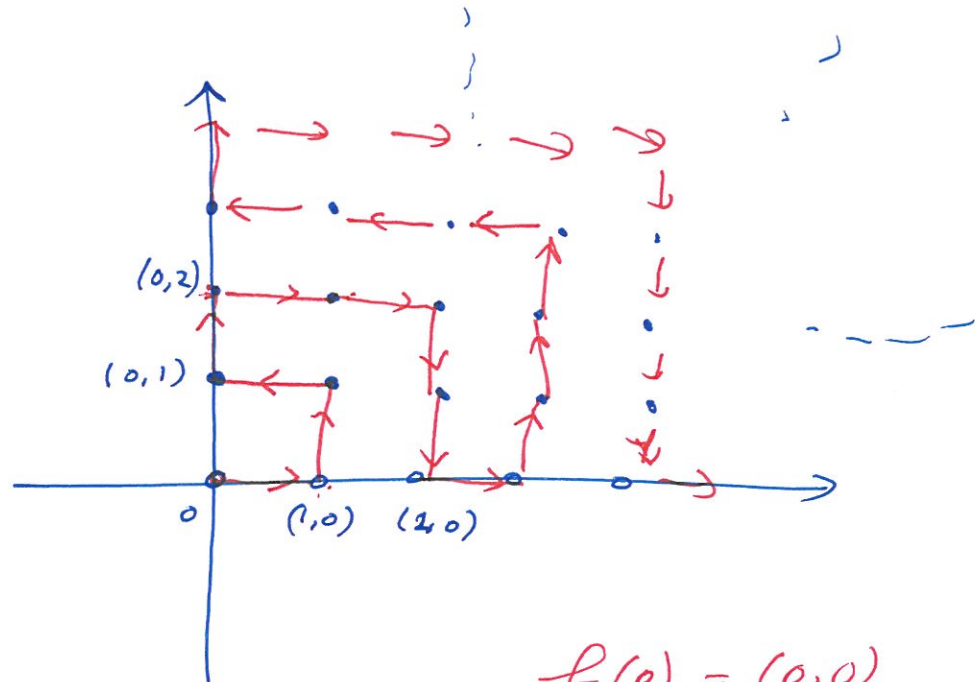
To prove $\mathbb{Z}$ is countable, we construct a bijection $f: \mathbb{N} \rightarrow \mathbb{Z}$

$$\begin{aligned} 0 &\mapsto 0 \\ 1 &\mapsto 1 \\ 2 &\mapsto -1 \\ 3 &\mapsto 2 \\ 4 &\mapsto -2 \\ &\vdots \\ n &\mapsto \end{aligned}$$

$$f(n) = \begin{cases} -\frac{n}{2} & \text{even} \\ \frac{n+1}{2} & \text{odd} \end{cases}$$



How about $\mathbb{N} \times \mathbb{N}$?



Theorem (Cantor, 1874): $\mathbb{R}$ is not countable.

Idea of the proof: We show $(0,1)$ is not countable.

0. 01011101011... -

Proof by Contradiction: Suppose $(0,1)$ is countable. Let us list the numbers.

~~0. $a_1^1 a_1^2 a_1^3 \dots$~~ ~~$a_i^j \in \{0,1\}$~~
~~0. $a_2^1 a_2^2 a_2^3 \dots$~~
 0. $a_1 a_2 a_3 a_4 \dots$ $a_i \in \{0,1\}$
 0. $b_1 b_2 b_3 b_4 \dots$ $b_i \in \{0,1\}$
 0. $c_1 c_2 c_3 c_4 \dots$ $c_i \in \{0,1\}$
 ⋮

We construct a number which is not in

this list $X = 0. x_1 x_2 x_3 x_4 \dots$

$$x_1 = \begin{cases} 1 & \text{if } a_1 = 0 \\ 0 & \text{if } a_1 = 1 \end{cases}$$

$$x_2 = \begin{cases} 1 & \text{if } b_2 = 0 \\ 0 & \text{if } b_2 = 1 \end{cases}$$